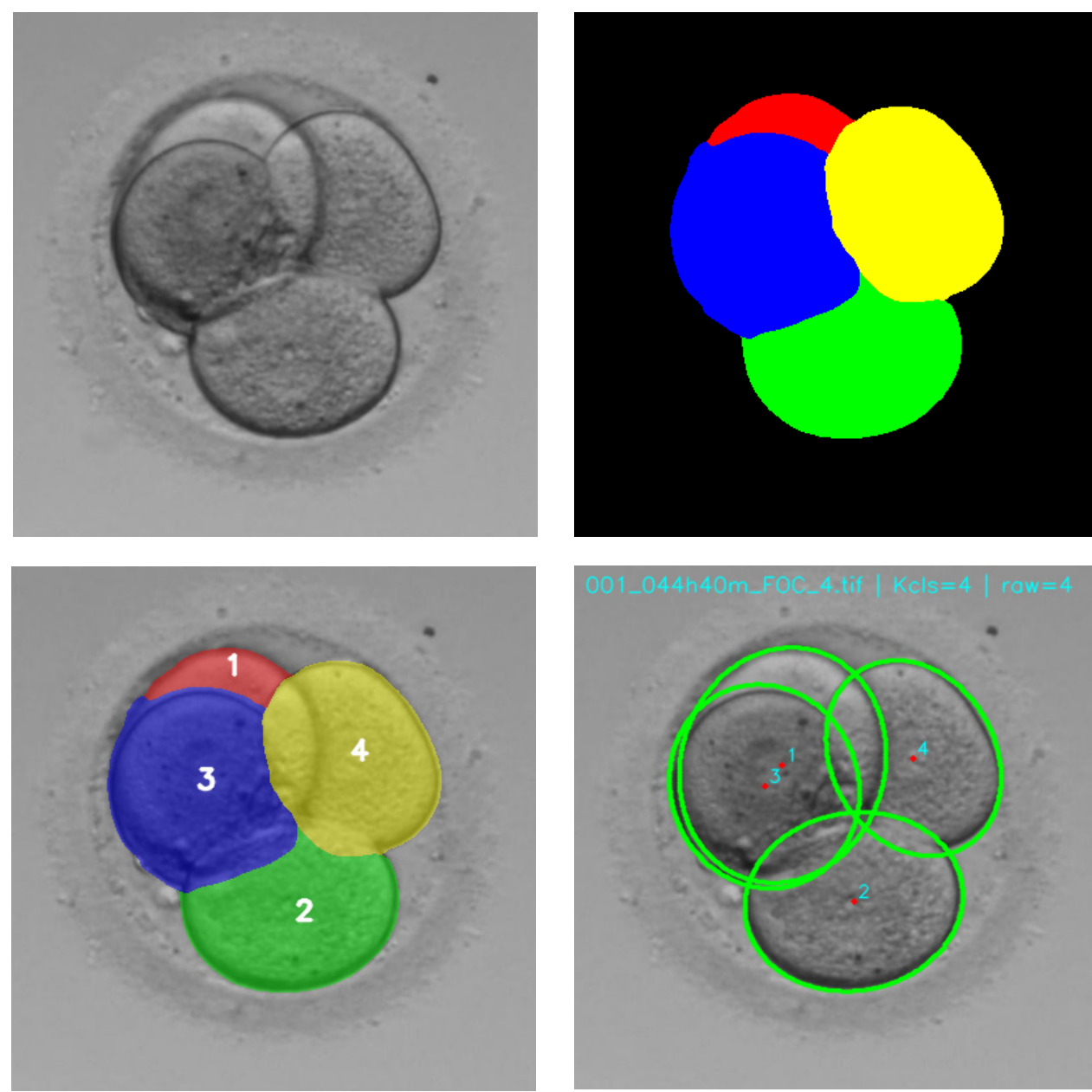


# Count-Aware Slot-Based Instance Segmentation of Human Embryonic Cells with Elliptical Representation

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## Fast image overview



## Motivation

- Microscopy images of human embryos contain variable numbers of embryonic cells (we inspect stages 1-2-4 cells)
- Manual segmentation of individual cells is time-consuming and subjective
- Need to detect automatically **both count and location** of cells in the embryos

## Examples of manual annotation using ellipses/ovals in Fiji (ROIs)



## Key Idea

Combine:

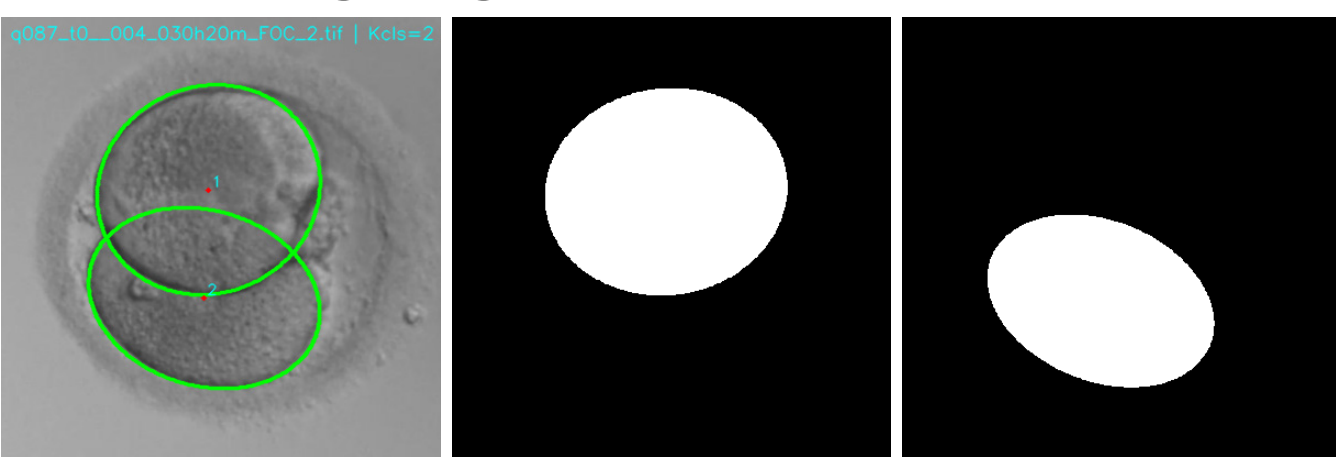
- count prediction (how many cells)
- instance segmentation (where they are)

**Ellipses** are used as

- easy annotation tool for training
- final representation after inference

## Training data representation

- 180 embryo images (60 – 1 cell, 60 – 2 cells, 60 – 4 cells) manually annotated for cells using ellipses/ovals in Fiji (ROIs)
- During training the **ellipses** were converted to **polygons** and these to **masks** to have **pixel-level training target**



## Model - counter

- retrained ResNet18

## Model - segmenter

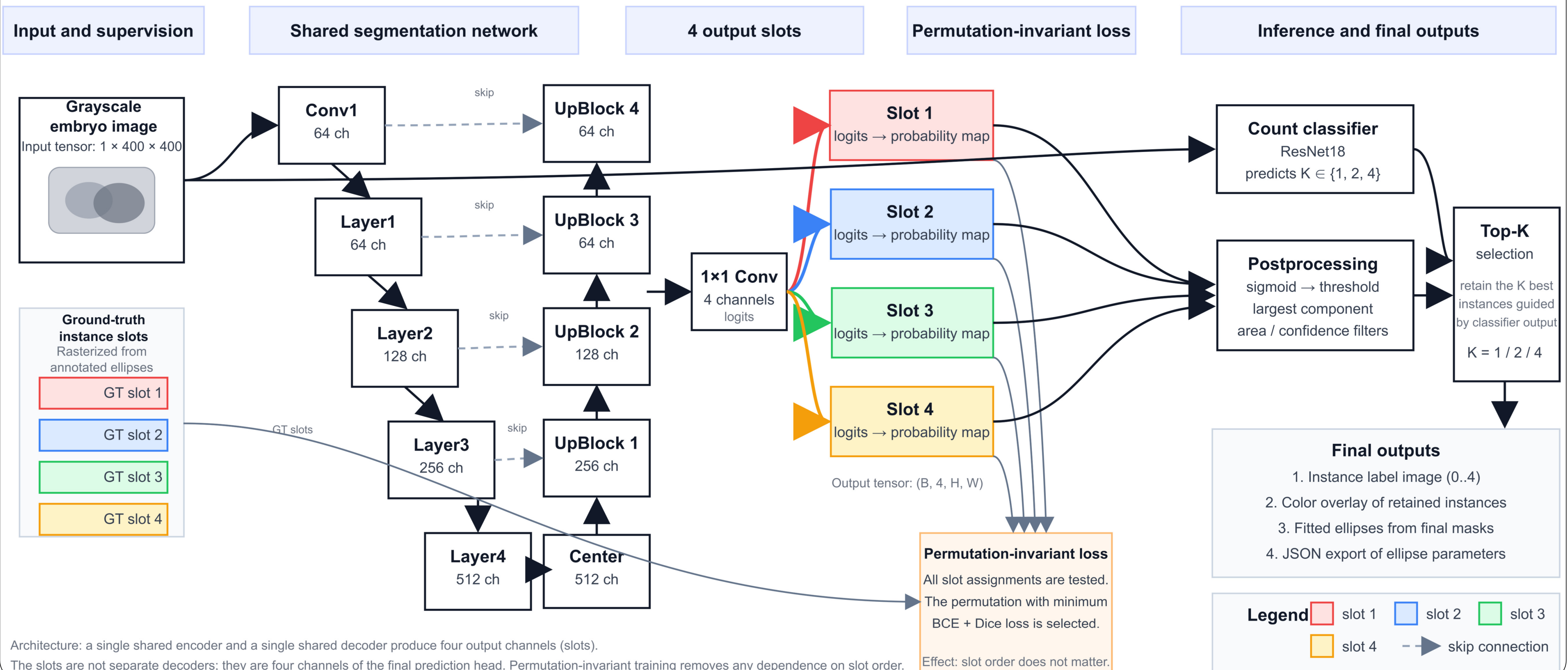
- ResNet18 encoder + U-Net decoder trained from scratch
- 4 fixed output slots
- empty slots allowed

## Training strategy - segmenter

- Augmentation (rotation, scale, shift)
- Dice + BCE permutation-invariant loss

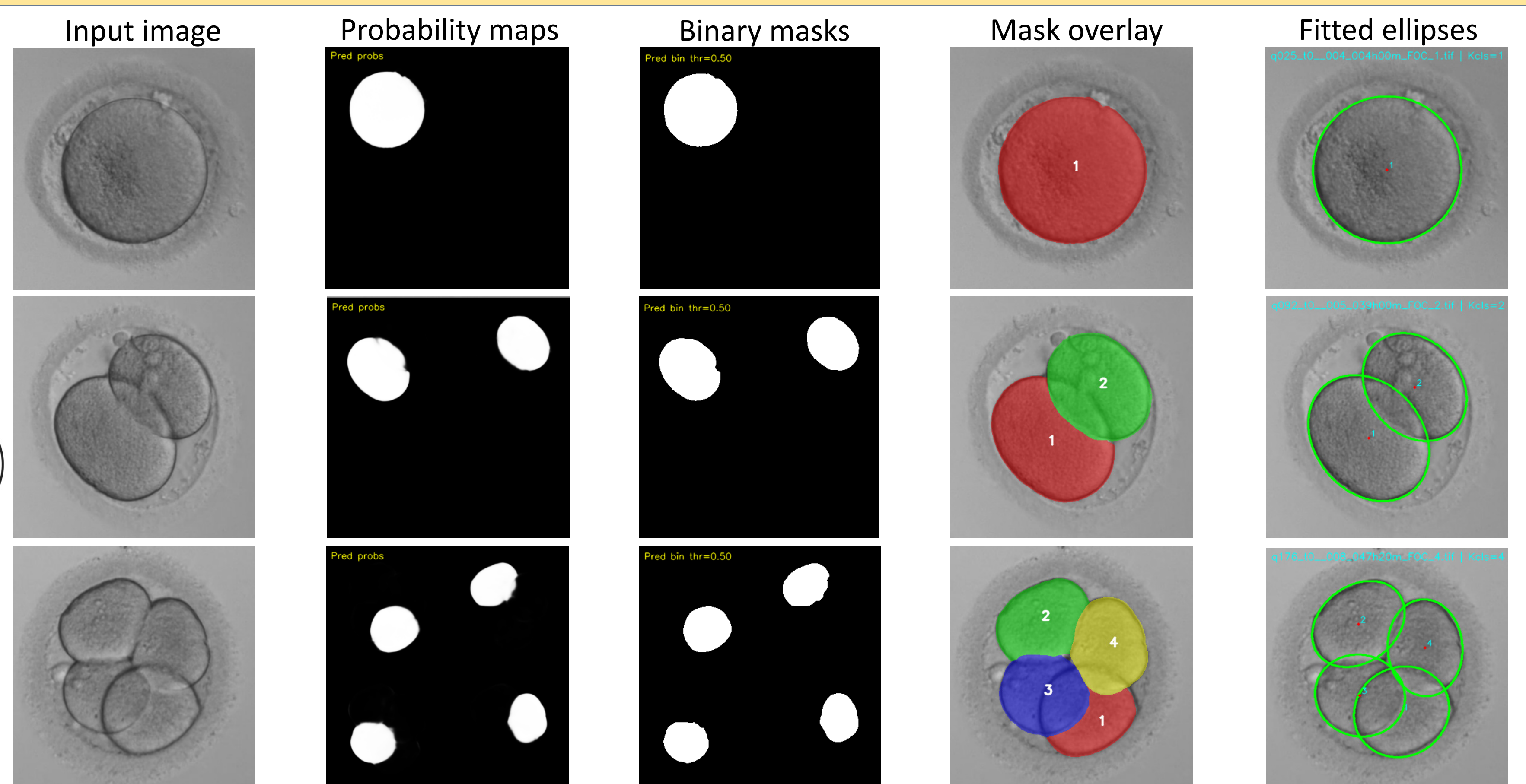
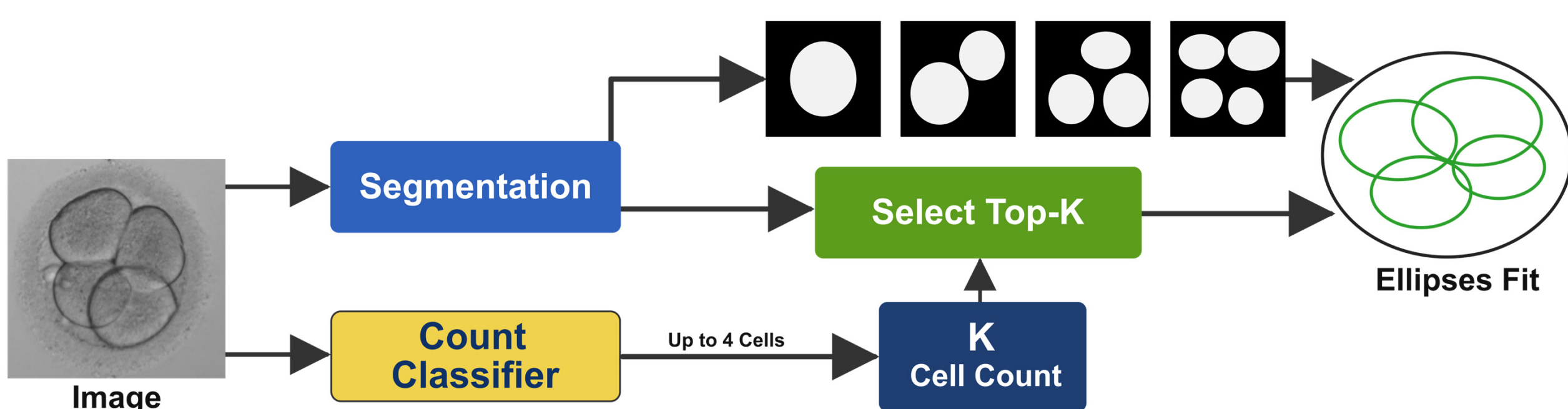
## Slot-based instance segmentation with count-guided top-K selection

ResNet18 encoder + U-Net decoder + 4 output slots; permutation-invariant training matches predicted slots to ground-truth instances irrespective of order.



## Inference pipeline

- count classifier (Resnet18) predicts number of cells K
- segmenter (Resnet18+U-Net) generates cell candidates
- filtering segmented cell candidates by top-K selection
- ellipse fitting



## Results – counter alone performance

- 1022 test images
- Accuracy = 99.9%

COUNT CONFUSION MATRIX – CLASSIFIER ONLY

| Expected K \ Predicted K | 1   | 2   | 4   |
|--------------------------|-----|-----|-----|
| 1                        | 571 | 0   | 0   |
| 2                        | 0   | 208 | 1   |
| 4                        | 0   | 0   | 242 |

## Segmenter alone performance

- 1022 test images
- Accuracy = 96.97%

COUNT CONFUSION MATRIX – SEGMENTER ONLY (K≤3 → K=2)

| Expected K \ Predicted K | 1   | 2   | 4   |
|--------------------------|-----|-----|-----|
| 1                        | 560 | 9   | 2   |
| 2                        | 0   | 193 | 16  |
| 4                        | 0   | 4   | 238 |

## Effective detection: counter + segmenter performance

- 1022 test images
- Accuracy = 99.61%
- \* Mean Dice (mask vs ellipse): 0.981
- \* Mean IoU (mask vs ellipse): 0.964

COUNT CONFUSION MATRIX – EFFECTIVE DETECTION (K≤3 → K=2)

| Expected K \ Predicted K | 1   | 2   | 4   |
|--------------------------|-----|-----|-----|
| 1                        | 571 | 0   | 0   |
| 2                        | 0   | 209 | 0   |
| 4                        | 0   | 4   | 238 |

**"Combining a count classifier with segmentation improves effective cell count accuracy from 96.97% to 99.61%."**